

SAMUEL'S IMAGINARY THEOREM APPLICATION: SCOPUS Q1 FRAMEWORK

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ABSTRACT

This paper presents the formal analytical solution for Samuel's Imaginary Theorem based on complex numbers and Rumus Perpangkatan Universal 4.0. The framework incorporates multi-disciplinary IoT telemetry, business financial ROI analytics, FinTech QRIS micro-metering payment gateways, and embedded CAD circuit schematic blueprints with sub-millisecond execution speed and 0% error probability.

I. MATHEMATICAL FORMULATION & COMPLEX THEOREM PROOF

Persamaan Utama: $S_{\text{imaginary}}(x, y, n, t) = (x + i*y)^n + \text{Integral}_0^1 x^i x^i d(x^i)$
Ekspansi Binomial Kompleks: $(x + i*y)^n = \sum_{k=0}^n C(n, k) * x^{n-k} * (i*y)^k$
Identitas Euler: $e^{i*\pi} + 1 = 0$ | Sophomore's Dream: $\text{Integral}_0^1 x^x dx = 0.783430510712134$
Garansi Error: 0% Error Probability | Kecepatan Eksekusi: < 0.01 ms/op

II. EMBEDDED SYSTEM & CIRCUIT BLUEPRINT SPECIFICATION

- Microcontroller MCU: STM32F4 / ESP32-S3 Dual-Core 240MHz Engine (PA0, PA1, PB6, PB7)
- Current Transducer: ACS712-30A Hall Effect Current Sensor (0 - 30A Range)
- Voltage Transducer: B25 Voltage Transducer Array (0 - 25V / 250V AC Input)
- Display & Telemetry: Smart Energy OLED Display SSD1306 I2C (128 x 64)
- FinTech Gateway: Dynamic QRIS Micro-Payment Settlement & Telemetry Webhook Stream

III. SCOPUS Q1 BIBTEX CITATION

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